Homework 0

William Flanders*
Haverford College
(Dated: January 25, 2023)

1. INTRODUCTION

My favorite equation:

You could have a citation to where this equation comes from

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0 \tag{1}$$

Equation 1 is the Black-Scholes Differential Equation. It might not necessarily be my favorite, but it is one that I love. It shows how an option's price relates to changes in the underlying stock and exemplifies how one could perfectly manage their portfolio to eliminate risk. I enjoy seeing equations that describe situations that are beyond what people normally expect. We tend to think of mathematics as only something to be used for physics, computer science, etc. In actuality, mathematics has influence in an extremely large number of disciplines including finance.

Nice job typesetting that equation. You clearly are comfortable using some of the most useful LaTeX commands.

One note is that equations are part of sentences so they should include any punctuation. In this case that would mean putting a period at the end of the equation.

Computational Physics/Astrophysics, Winter 2023:

Grading Rubrics [1]

Haverford College, Prof. Daniel Grin

For coding assignments, roughly 25 points will be available per problem.

1. Does the program complete without crashing in a reasonable time frame? If yes, up to +3 points.

2. Does the program use the exact program files given (if given), and produce an answer in the specified format? If yes, +1 points

3. Does the code follow the problem specifications (i.e numerical method; output requested etc.) Up to +2 points

4. Is the answer correct? Up to+4points

5. Is the code readable? Up to+2points

. 5.1. Are variables named reasonably?

. 5.2. Are the user-functions and imports used?

. 5.3. Are units explained (if necessary)?

. 5.4. Are algorithms found on the internet/book/etc. properly attributed?

[1] Inspired by rubric of D. Narayanan, U. Florida, and C. Cooksey, U. Hawaii

6. Is the code well documented? +3points

. 6.1. Is the code author named?

. 6.2. Are the functions described and ambiguous variables defined?

. 6.3. Is the code functionality (i.e. can I run it easily enough?) documented?

7. LaTeX writeup (up to 10 points)

- Are key figures and numbers from the problem given? (3 points)
- +2 Is a brief explanation of physical context given? (2 points)
- +1 If relevant, are helpful analytic scalings or known solutions given? (1 point)
- +2 (only needed 1 for this assignment) Are 3-4 key equations listed (preferably the ones solved in the programming assignment) and algorithms named? (2 points)
- +1 Are collaborators clearly acknowledged? (1 point)
- Are any outside references appropriately cited? (1 point)

+1 (no citation for an equation that is not common knowledge. No points off now but you should cite in the future.)

Note, even if (1), (2), (3), or (4) are not correct, one can still obtain many points via (5), (6), and (7).